Disaster-Response Wi-Fi Beacon Locator

AI + IoT + Edge Computing + Drone-Assisted Search & Rescue

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Project: HillGuard App & Wi-Fi Locator Ecosystem



1. Introduction to the Technical Framework

The technical architecture of the HillGuard Disaster-Response Wi-Fi Beacon Locator is a comprehensive, multi-layered system designed to operate in extreme, infrastructure-denied environments. By bridging the gap between passive wireless signals and actionable rescue intelligence, the system employs a sophisticated integration of IoT sensing, Edge AI, and Drone-Assisted Search & Rescue (UAS).

The core philosophy of our technical approach is strictly analytical: we utilize Machine Learning (ML) as a decision-support tool to rank and prioritize search areas, explicitly acknowledging that ML does not definitively confirm survivor presence. Final operational maneuvers and route execution remain operator-approved, adhering to stringent local safety rules. This report details the five primary technical phases of our solution: Data Sources, Data Processing, Feature Engineering, ML Search-Priority Modeling, and the Drone-Assisted Rescue ecosystem.

2. Data Sources and Acquisition Layer

The foundation of the HillGuard system relies on a multi-modal data acquisition strategy. By aggregating heterogeneous data types, the system mitigates the inherent noise and multipath fading associated with disaster environments.

2.1 Passive Wi-Fi Signal Sensing

The primary localization vector utilizes permitted Wi-Fi transmissions emitted by standard consumer devices (e.g., smartphones, wearables). When cellular networks fail, these devices continuously emit "probe requests" to discover known networks. Our distributed edge nodes passively capture these 802.11 frames. The critical data points extracted include:

- ***Received Signal Strength Indicator (RSSI):*** *Used as the primary metric for distance bounding and spatial triangulation.*
- ***Timestamps:*** *Microsecond-precision logs essential for chronological synchronization across multiple receiving nodes.*
- ***MAC Metadata:*** *Hashed or anonymized identifiers to track signal persistence without compromising user privacy.*

2.2 IoT Environmental Sensors

To contextualize the wireless data, the system integrates localized IoT sensor arrays. These sensors provide vital environmental telemetry:

- ***Temperature and Humidity:*** *Helps in assessing the survivability of a specific micro-environment and calibrates thermal imaging expectations.*
- ***Device Status & Power:*** *Monitors the health, battery life, and operational status of the scanning nodes themselves.*
- ***Motion and Water Levels (Optional):*** *Provides real-time hazard context (e.g., rising floodwaters or structural shifting) to safeguard rescue teams.*

2.3 GPS and Ground Node Telemetry

Accurate spatial anchoring is paramount. The system continuously logs:

- ***Receiver Coordinates:*** *High-precision GNSS/GPS data from static ground nodes.*
- ***Observation Time:*** *Temporal anchoring to correlate moving signals with static observations.*

2.4 Unmanned Aerial System (UAS) Observations

Drones serve as dynamic data collection nodes, extending the sensory reach of the system into inaccessible areas.

- ***Aerial Imagery & Video:*** *High-resolution RGB feeds for topographical mapping and visual anomaly detection.*
- ***Thermal Sensing (Optional):*** *Infrared thermography to detect heat signatures in low-light or obscured environments.*

3. Data Processing and Edge Integration

Raw telemetry collected in a disaster zone is highly erratic. Our processing pipeline operates at the edge, ensuring continuous functionality even during total internet blackouts.

3.1 The 4-Step Processing Pipeline

1. ***Collect:*** *Uninterrupted aggregation of wireless frames, sensor telemetry, and aerial feeds into a local volatile buffer.*
2. ***Clean & Validate:*** *Algorithmic filtering of anomalous data. This includes dropping malformed Wi-Fi packets, smoothing erratic RSSI spikes caused by immediate multipath reflections, and removing duplicate probe requests captured within the same microsecond window.*
3. ***Synchronize Time + Coordinates:*** *The most critical step in distributed sensing. Network Time Protocol (NTP) or localized RTCs (Real-Time Clocks) are used to synchronize timestamps across all ground nodes and drones. GPS coordinates are fused with these synchronized timestamps to create a spatio-temporal matrix of the disaster zone.*
4. ***Extract Features:*** *Transforming the cleaned raw data into mathematically structured inputs suitable for Machine Learning ingestion.*

3.2 Edge / Command Platform Architecture

Relying on cloud infrastructure during a catastrophe is a fundamental flaw in traditional systems. The HillGuard architecture utilizes a resilient Edge Platform.

- ***Local Buffering (Offline-First):*** *All data is stored locally on the edge nodes during network outages. The system is designed to operate autonomously, calculating initial priorities without cloud intervention.*
- ***Secure Data Transfer:*** *When an ad-hoc network or satellite backhaul becomes available, the edge nodes synchronize their databases with the central command securely.*

- ***Operator Dashboard:*** *A locally hosted, web-based interface allowing field commanders to view processed data and ML outputs instantly, without waiting for cloud rendering.*

4. Feature Engineering

The effectiveness of our spatial ML model relies entirely on the quality of engineered features. We transform raw variables into contextual indicators of human presence.

4.1 RSSI Patterns Over Time

Rather than utilizing single RSSI snapshots, we engineer time-series features. We calculate the moving average, variance, and standard deviation of RSSI values from specific anonymized devices. A highly variable RSSI might indicate a moving survivor, while a stable, low-variance RSSI suggests a trapped individual or a stationary dropped device.

4.2 Signal Observation Count

The frequency of intercepted signals is a strong indicator of priority. Devices that transmit repeatedly from a constrained spatial area over an extended timeframe generate a high "observation count" feature, elevating the likelihood of a trapped survivor versus a transient signal (e.g., someone fleeing the area).

4.3 Sensor and Device Status Context

We engineer features that combine environmental danger with signal presence. For instance, if a Wi-Fi signal is detected in a zone where IoT sensors report rapidly rising water levels or extreme temperatures, the algorithm applies a hazard multiplier to the priority score.

4.4 GPS / Drone Position Correlation

The relative distance vector between moving drones and static Wi-Fi signals is calculated. As a drone maneuvers, it acts as a dynamic node, creating a synthetic aperture of signal readings. This feature maps the intersection of ground node radii with drone observation paths.

4.5 Visual Scene Cues and Uncertainty Metrics

- ***Visual Cues:*** *Metadata tags generated from the drone's visual survey (e.g., "rubble density," "blocked path") are converted into categorical features.*
- ***Signal and Location Uncertainty:*** *We engineer a specific feature representing standard error. In areas with high signal noise (dense concrete environments), the uncertainty feature increases, which the ML model uses to expand the geographical radius of the priority zone rather than incorrectly pinpointing a false coordinate.*

5. Machine Learning Search-Priority Model

The core analytical engine of HillGuard is a tailored Machine Learning model designed not to locate a precise coordinate, but to rank macroscopic geographical zones based on the probability of successful rescue.

5.1 Explainable Baseline: Random Forest Classification

We utilize a Random Forest (RF) ensemble learning algorithm as our primary model. The selection of RF is deliberate and backed by in-depth research into disaster response constraints:

- ***Explainability:*** *Unlike deep neural networks, Random Forests allow operators to inspect feature importance. If a zone is ranked "High Priority," the operator can see exactly why (e.g., high RSSI count + stable signal).*
- ***Non-Linearity & Robustness:*** *RF excels at handling the non-linear relationship between RSSI decay and physical distance, especially in environments heavily distorted by rubble and structural collapse.*
- ***Edge Efficiency:*** *Once trained, RF inference is computationally inexpensive, allowing it to run rapidly on localized edge hardware without draining battery reserves.*

5.2 Model Outputs: Ranking, Confidence, and Uncertainty

The RF model outputs a multi-dimensional array for candidate zones:

1. ***Rank (High/Medium/Low):*** *A categorical prioritization of search areas.*
2. ***Confidence Score (0.0 to 1.0):*** *A probabilistic output indicating the model's certainty. A score of 0.92 indicates strong multi-node corroboration, while 0.34 indicates a weak or single-node observation.*
3. ***Uncertainty Area:*** *The model dynamically adjusts the bounding box of the search zone. High uncertainty results in a broader search perimeter.*

5.3 Validation Metrics

To ensure operational readiness, the model is strictly evaluated against predefined metrics during field testing:

- ***Detection Rate (Recall):*** *The percentage of actual devices present that the system successfully flags for investigation.*
- ***False-Alarm Rate:*** *Minimizing "ghost" signals to prevent the misallocation of critical rescue resources.*
- ***Localization Error:*** *Measured in meters, representing the distance between the model's estimated centroid and the actual device location.*
- ***Latency:*** *The time elapsed from signal capture to prioritization output on the dashboard, ensuring rapid actionable intelligence.*

6. Drone-Assisted Search & Rescue Integration

The aerial component transforms the 2D ground intelligence into a 3D tactical operation. The UAS integration follows a strict, autonomous-capable operational loop.

6.1 The Survey -> Observe -> Report Workflow

1. ***SURVEY:*** *Directed by the ML priority map, drones are dispatched to inspect inaccessible, structurally compromised, or hazardous terrains. They operate on operator-approved waypoints to ensure airspace safety.*

2. ***OBSERVE:*** *While traversing the route, the drone utilizes its payload. It captures high-resolution aerial imagery, video, and (if equipped) thermal data. It simultaneously acts as a mobile Wi-Fi scanner, adding z-axis signal strength data to the ground matrix.*
3. ***REPORT:*** *The drone continuously streams or periodically bursts (depending on network availability) its observations back to the Command Dashboard. This allows human operators to review the visual and thermal evidence against the wireless signal heatmaps.*

6.2 Specialized Classification: Human vs. Animal Rescue

A unique capability of the HillGuard aerial system is its visual and thermal classification logic, designed to optimize resource allocation:

- ***Human Rescue:*** *Edge-based Computer Vision algorithms process drone imagery to flag distinct human silhouettes, clothing anomalies against rubble, or specific distress cues (e.g., waving, SOS patterns).*
- ***Animal Rescue:*** *The thermal and visual signatures of livestock or pets differ significantly from humans. The system identifies possible trapped or stranded animals, allowing commanders to dispatch specialized animal rescue teams rather than misdirecting human medical personnel.*

7. Command Dashboard & Decision Support

The culmination of this technical architecture is the Command Dashboard—a centralized, live operational picture.

7.1 Visualizing the Intelligence

The dashboard overlays several critical data layers onto a topographical map:

- ***Wireless Signal Heatmap:*** *A visual gradient (Red/High to Blue/Low) showing the raw detected wireless signals and their estimated range (e.g., Strongest Signal: -48 dBm, Estimated Range: 80–120 m).*
- ***Candidate Priority Zones:*** *Distinct, color-coded polygons (Red/Orange/Green) outlining the ML-driven search areas with their associated confidence scores.*
- ***Drone Telemetry:*** *Real-time plotting of the Drone Route, Waypoints, Scanned Areas, and live Drone Position.*
- ***Target Indicators:*** *Specific visual pings highlighting "Possible Human Detected" alongside the vision model's confidence rating.*

7.2 The Operator-in-the-Loop Protocol

Despite the advanced AI and autonomous drone routing, the technical approach mandates human oversight. The system serves purely as ***Decision Support****. The ML ranks areas to investigate; it does not explicitly confirm survivor presence. All drone routes, flight operations, and rescue team deployments remain strictly operator-approved and subject to local safety rules, ensuring ethical, safe, and effective disaster response.*